

# Ethanol Blending in Gasoline – Germany

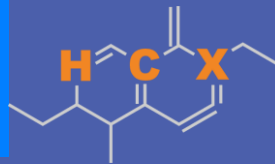
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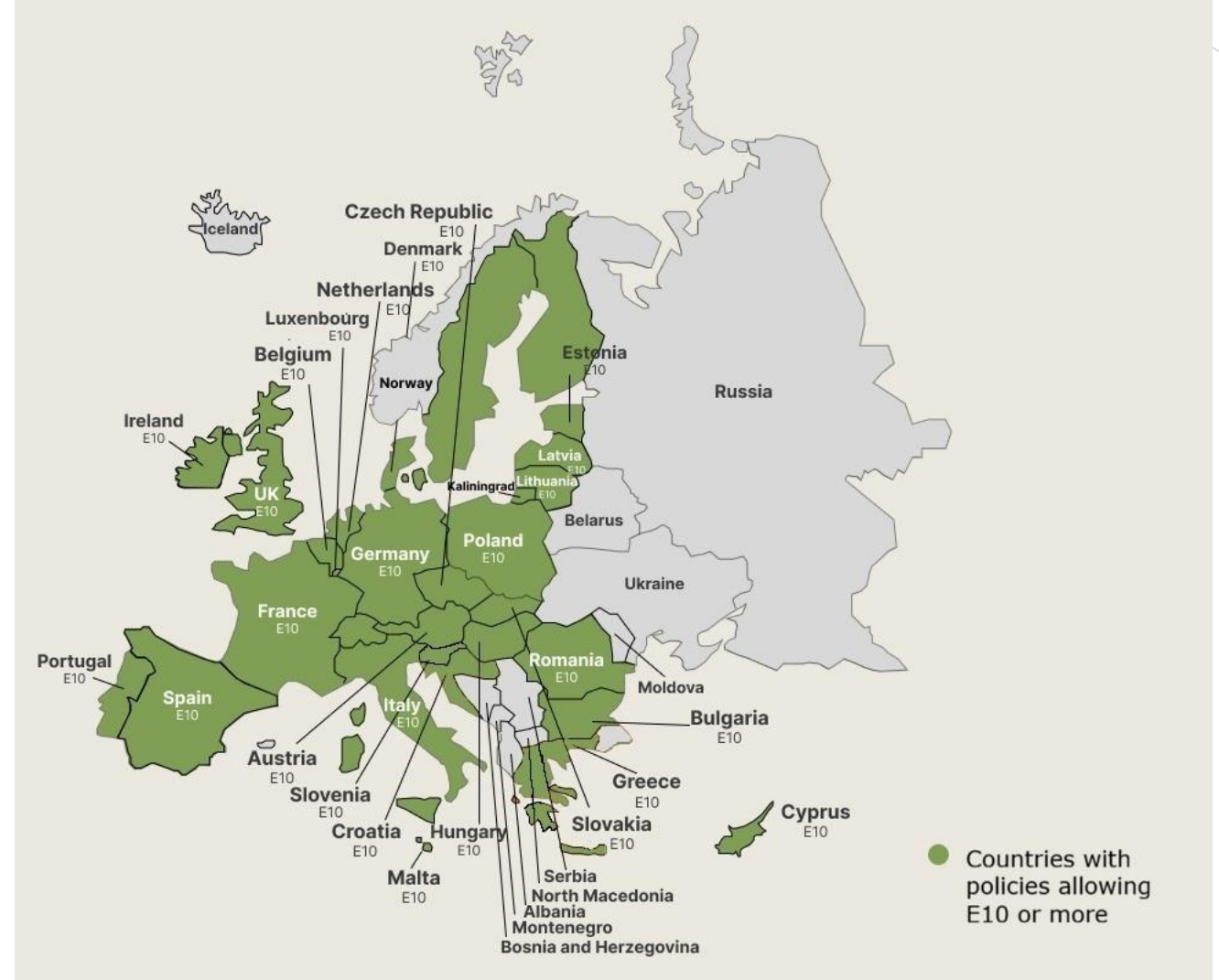


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# Ethanol Blending in Europe

There are important fuel quality and environmental impact of vehicle emission challenges in the Region.

- The use of ethanol improves gasoline quality and creates flexibility in gasoline production.
- Ethanol use is a cost-effective way to increase gasoline octane and to replace more expensive gasoline components.
- Ethanol contributes to transport decarbonization and air quality improvement.
- There are opportunities across Europe to increase the ethanol blend level and implement new policies on the use of gasoline-ethanol blends.



Twenty eight countries with potential and additional use of ethanol were studied: 1) Gasoline market profiles; 2) Optimization of gasoline blends with ethanol and, 3) Environmental impact of gasolines blended with ethanol.

# Ethanol Gasoline Blending in Germany



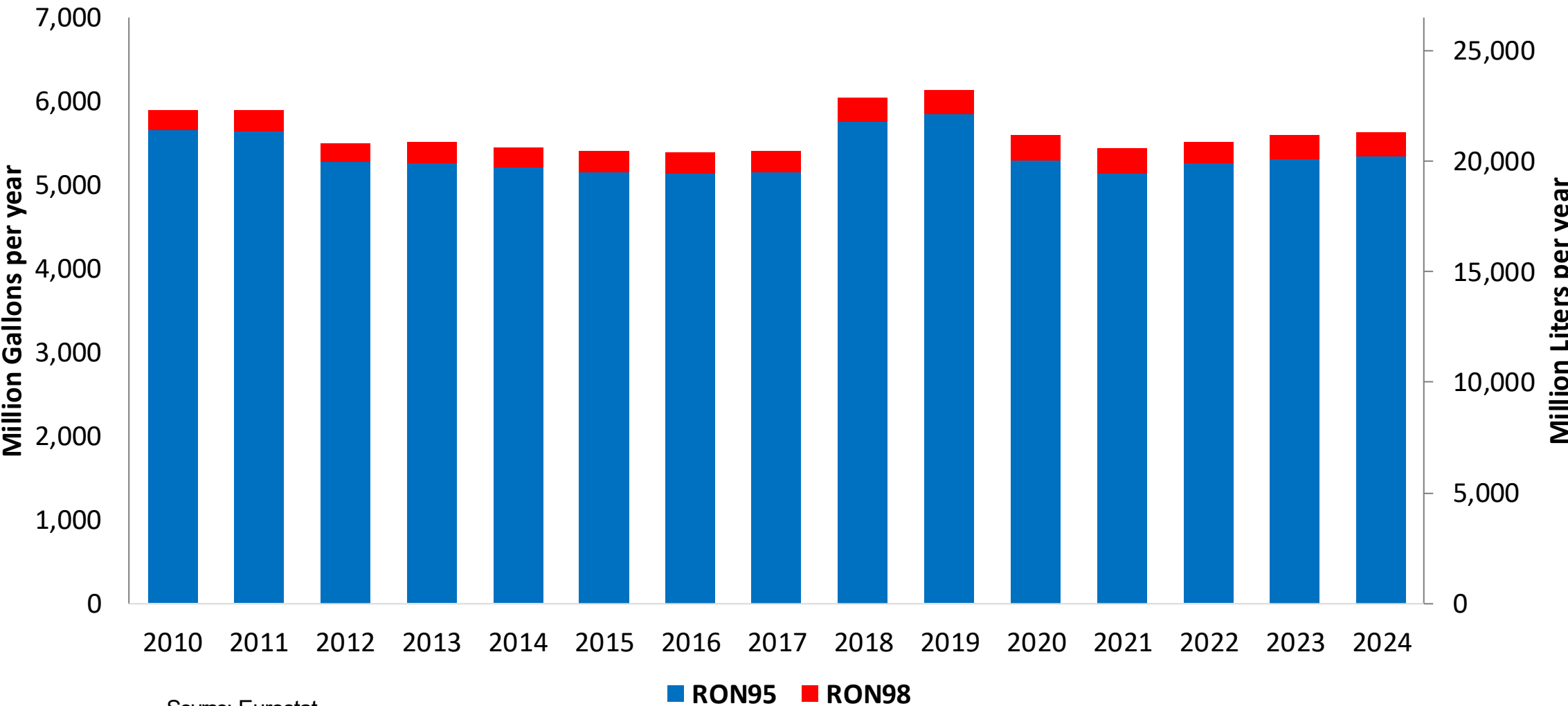
In 2024, gasoline consumption in Germany was 21,292 million liters (5,625 million gallons) and growth rate was -1.7%. Gasoline production and net imports were 22,284 and -3,611 million liters (5,887 and -954 million gallons) correspondingly. Germany complies with the European gasoline standards and offers two main grades: RON 95 and RON 98 with an average octane of 95.2 RON.

In 2024, ethanol consumption in Germany was 2,262 million liters (598 million gallons) representing 10.6% of gasoline demand. Ethanol production and Net Imports were 1,333 and 929 million liters (598 and 245 million gallons) correspondingly. Ethanol sources are Cereals 64% and Sugar Beet 33%.

Source: Eurostat, European Commission

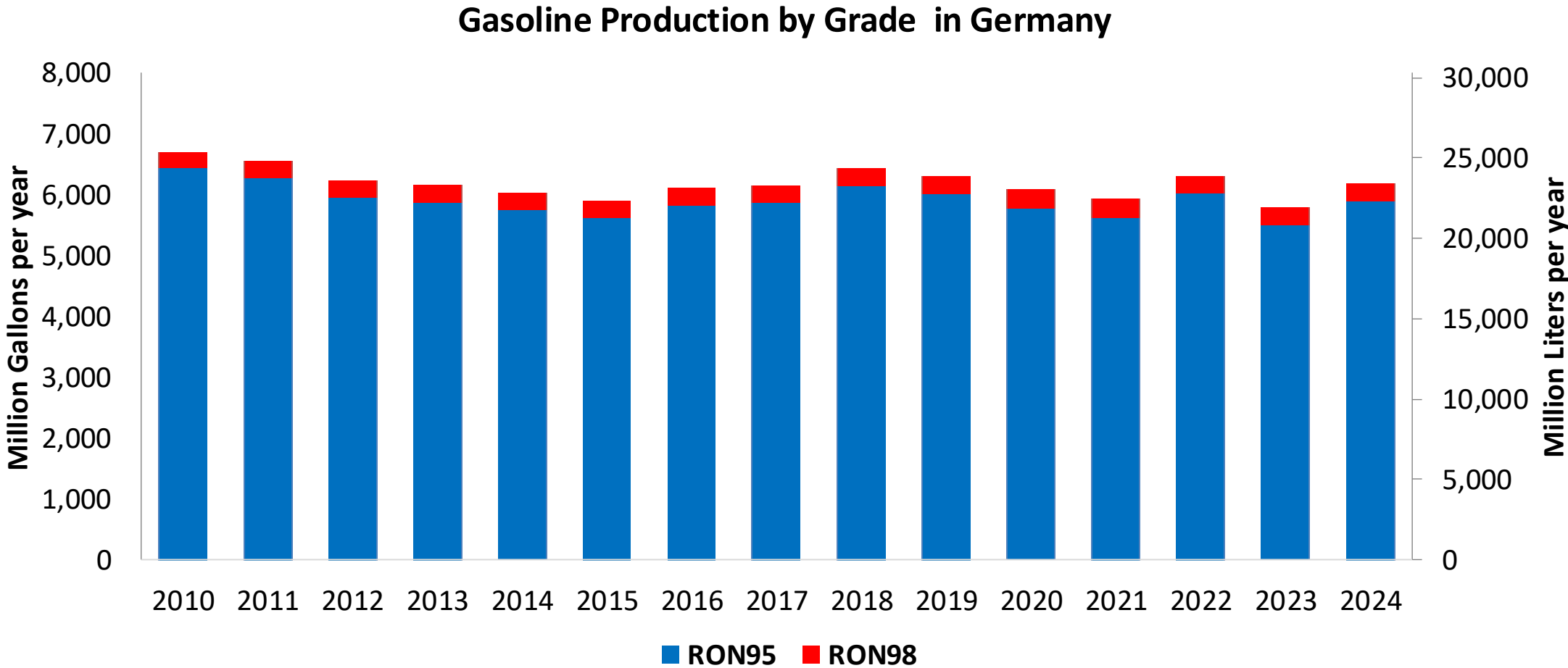
# Gasoline Demand in Germany

Gasoline Demand by Grade in Germany



Source: Eurostat

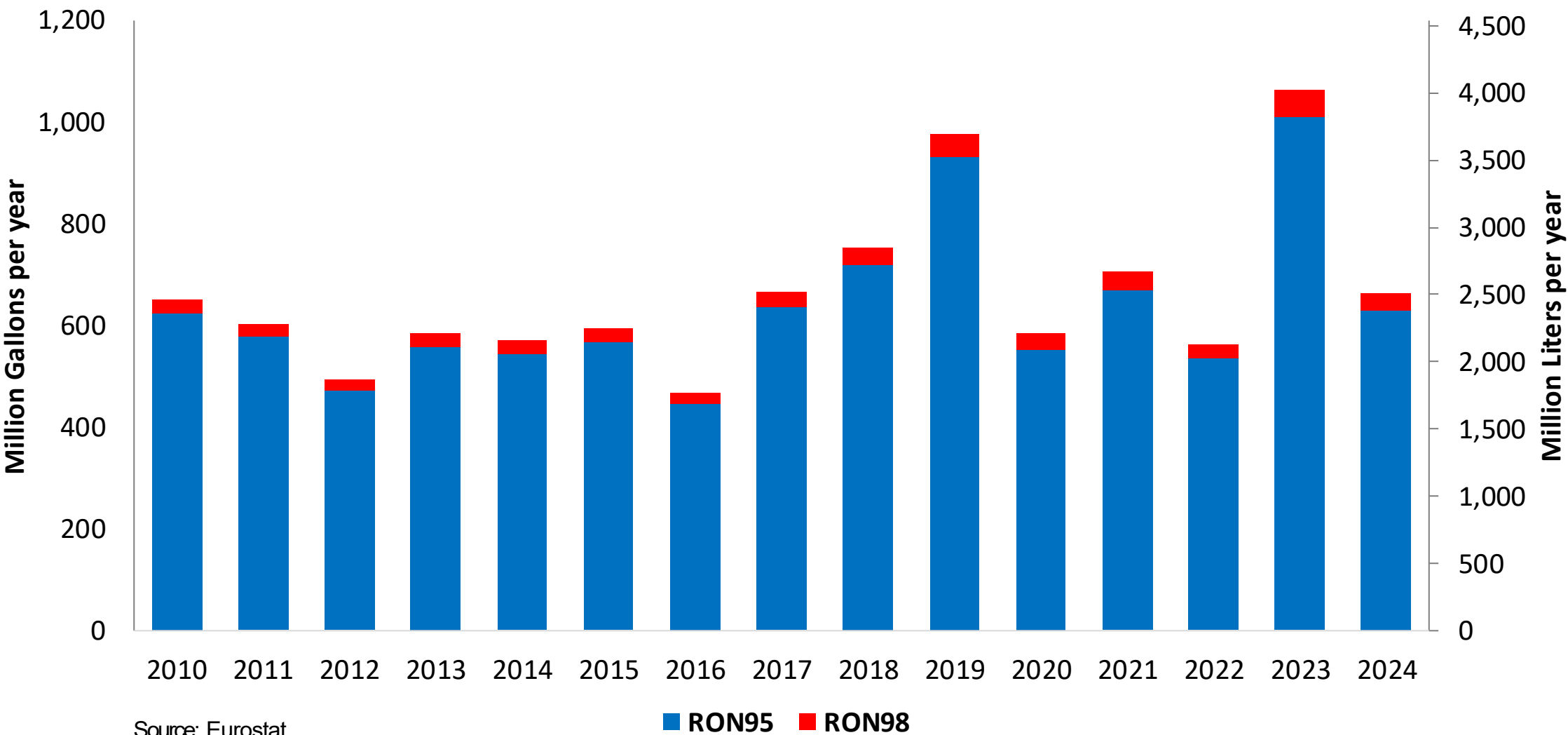
# Gasoline Production in Germany



Source: Eurostat

# Gasoline Imports in Germany

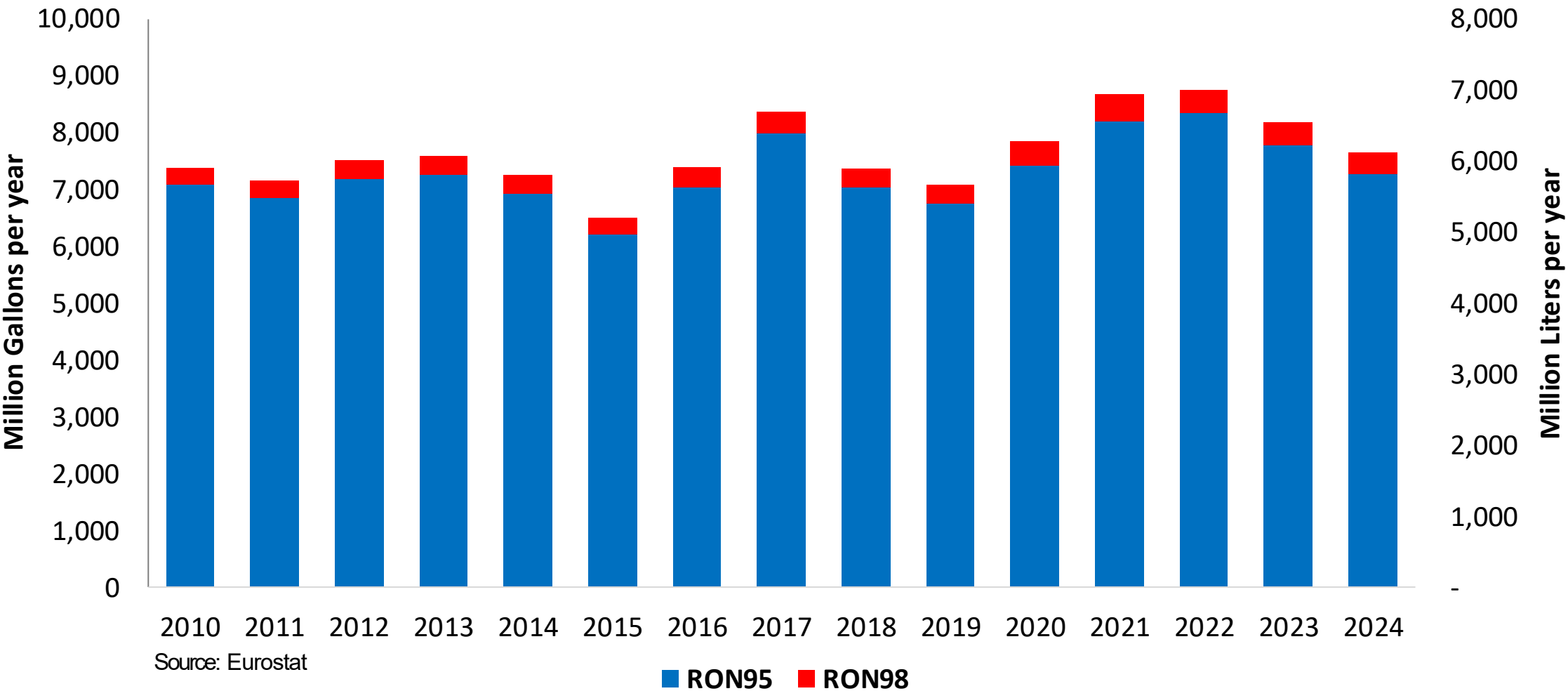
Gasoline Imports by Grade in Germany



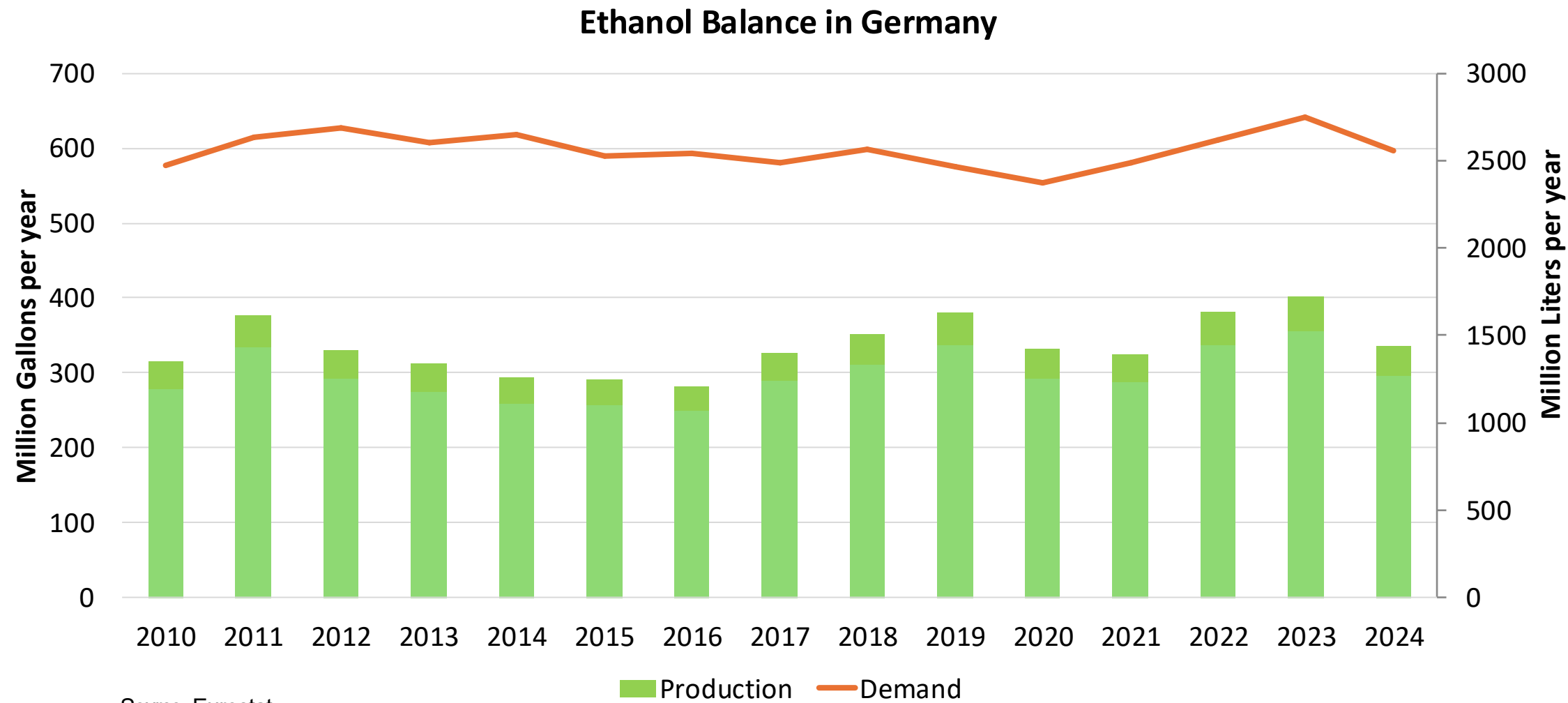
Source: Eurostat

# Gasoline Exports in Germany

Gasoline Exports by Grade in Germany



# Ethanol Balance in Germany





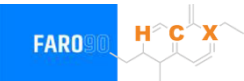
# Gasoline Quality in Germany

| Name                                | EN 228:2025 (Euro 6); Country Level Fuel Ordinances |       |            |       |           |       |            |       |
|-------------------------------------|-----------------------------------------------------|-------|------------|-------|-----------|-------|------------|-------|
| Implementation Date                 | 2025                                                |       |            |       |           |       |            |       |
| Applicability                       | All EU Contries + UK                                |       |            |       |           |       |            |       |
| Grade                               | RON 95 E5                                           |       | RON 95 E10 |       | RON 98 E5 |       | RON 98 E10 |       |
|                                     | Min                                                 | Max   | Min        | Max   | Min       | Max   | Min        | Max   |
| RON                                 | 95.0                                                |       | 95.0       |       | 98.0      |       | 98.0       |       |
| MON                                 | > 85                                                | > 88  | > 85       | > 88  | > 85      | > 88  | > 85       | > 88  |
| AKI                                 |                                                     |       |            |       |           |       |            |       |
| Benzene (%vol)                      | < 1                                                 |       |            |       |           |       |            |       |
| Aromatics (% vol)                   | < 35                                                |       |            |       |           |       |            |       |
| Olefins (%vol)                      | < 18                                                |       |            |       |           |       |            |       |
| Lead (g/L)                          | < 0.005                                             |       |            |       |           |       |            |       |
| Manganese (mg/L)                    | < 2,0                                               |       |            |       |           |       |            |       |
| Sulfur (ppm)                        | 10.0                                                |       |            |       |           |       |            |       |
| Oxygen (%w)*                        | < 2,7                                               | < 3,7 | < 2,7      | < 3,7 | < 2,7     | < 3,7 | < 2,7      | < 3,7 |
| Ethanol (%vol)                      | < 5                                                 | < 10  | < 5        | < 10  | < 5       | < 10  | < 5        | < 10  |
| PVR 37.8°C (psi)                    | <> 45 - 100 kPa in 6 Volatility Classes             |       |            |       |           |       |            |       |
| Overall Biofuels (%cal)             | -                                                   |       |            |       |           |       |            |       |
| Advanced Biofuels (%cal)            | 0.7                                                 |       |            |       |           |       |            |       |
| Bioethanol in Gasoline Sales (%cal) | -                                                   |       |            |       |           |       |            |       |
| GHG Emission Reduction (%)          | 10.5                                                |       |            |       |           |       |            |       |

\* Methanol 3%, Iso-propyl alcohol 12%, Ter-butyl alcohol 15%, Iso-butyl alcohol 15%, Ethers +5 Carbons 22%, Other Oxygenates 15%

Source: European Commision

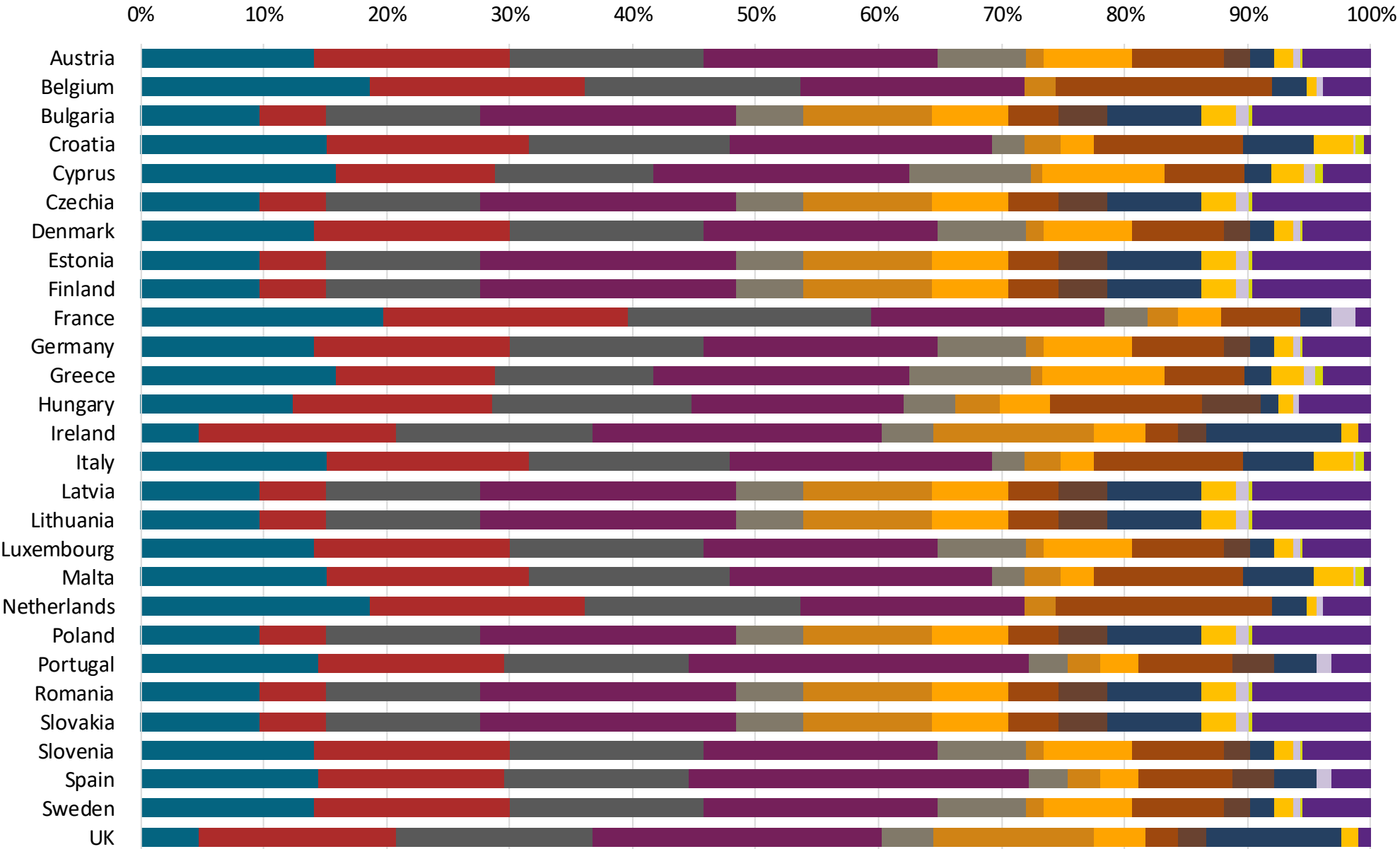
# Gasoline Component Blending in Europe



Gasoline is a blend of a base gasoline and other components. Each component provides different properties to the final blend, for example, isomerates, alkylates and butanes increase the octane. The components commonly used in Europe are:

- Catalytic Gasoline
- Light Primary Naphta
- Heavy Primary Naphta
- Reformate
- Isopentane
- Alkylate
- Isomerate
- Hydrocracked Gasoline
- Coker Naphta
- n-Butane
- MTBE
- ETBE
- TAME
- Aromatics

Source: Faro90



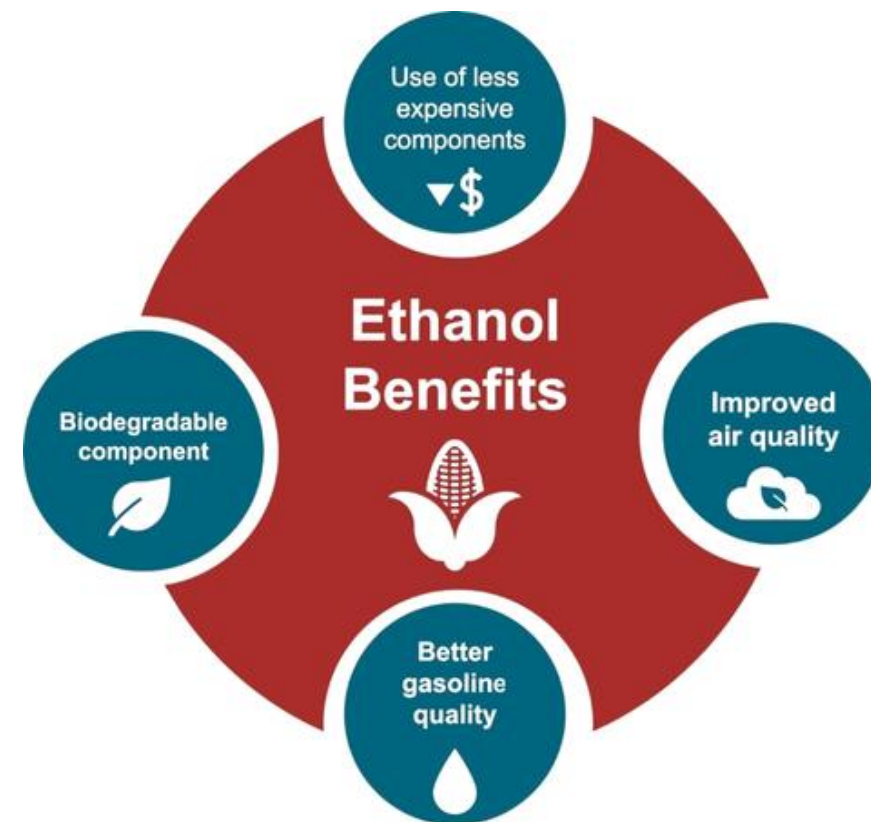
# Gasoline Blending Optimization

In some parts of the world, ethanol is added to gasoline as a blending component. The advantages of ethanol include that it is a renewable fuel made of biomass; that it is an octane booster that helps to dilute sulfur; and that it allows the fulfillment of environmental objectives. To determine the optimal components to be blended with ethanol, a **blending model** was used. This model selects the components to add in the gasoline/ethanol blend based on:

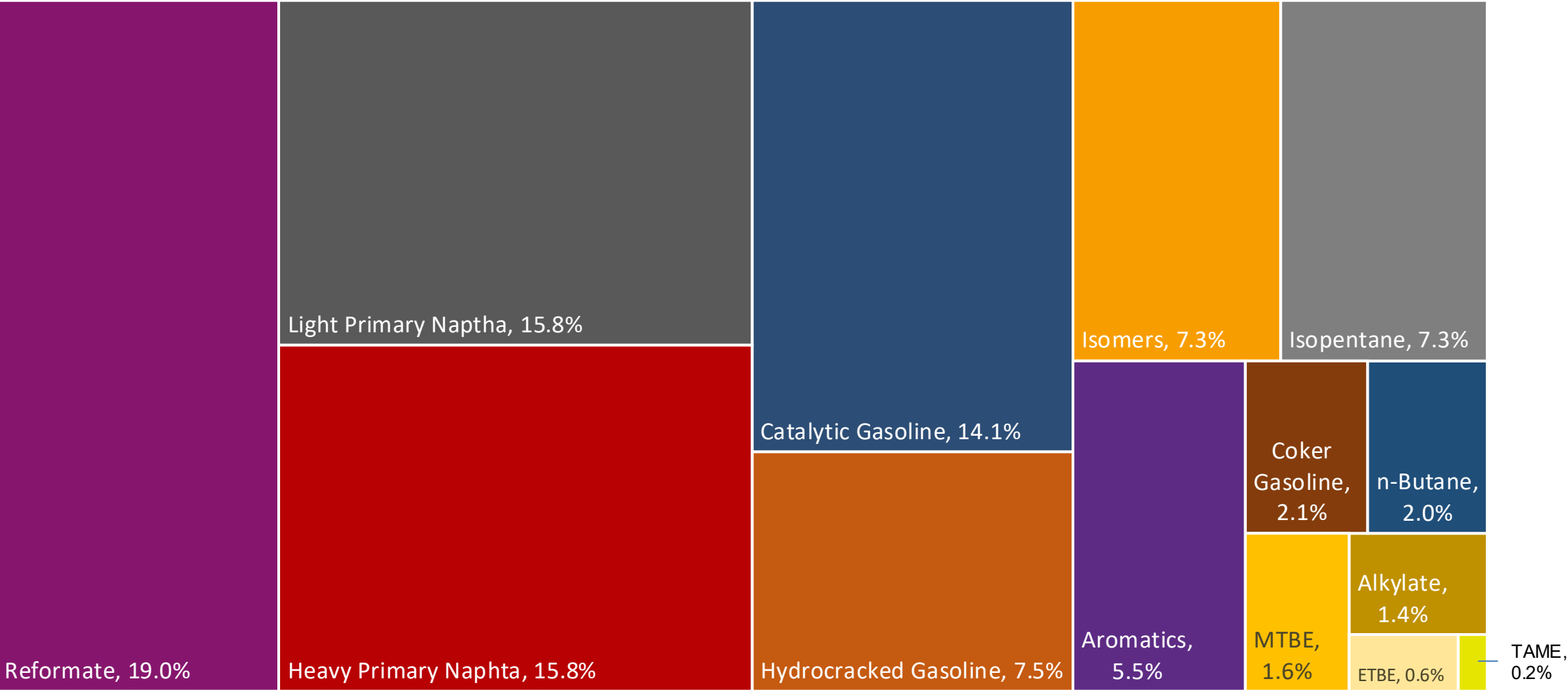
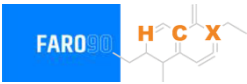
- Components prices,
- Properties each component affects,
- Quality parameters by country, and
- Component availability by country.

Through iterations, the model obtains the %v/v of the components to be blended with 10%, 15%, 20%, 25% and 30% of ethanol, in such a way that the final blend complies with the required properties of a finished gasoline by country.

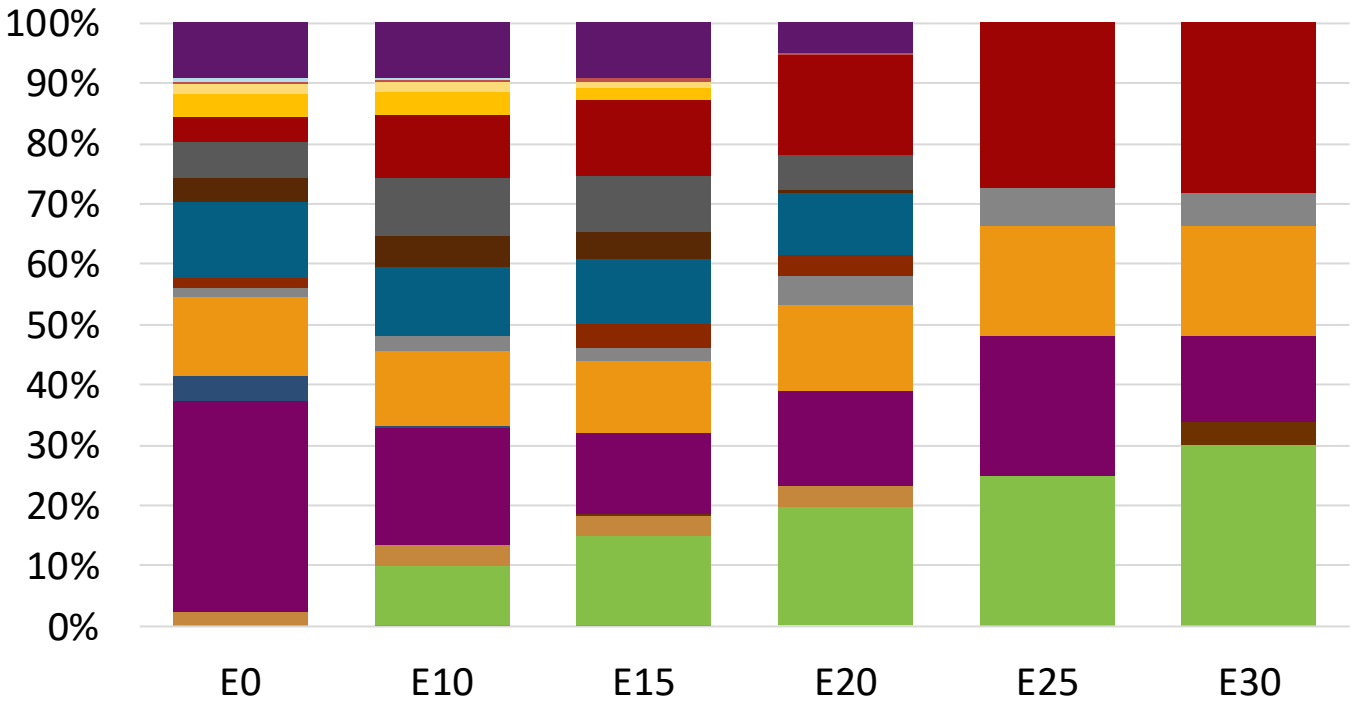
The blending model uses gasoline component spot average prices 2024 and provides fuel prices that do not include country distribution costs, local taxes and subsidies and import or gas station margins.



# Germany Available Blending Components



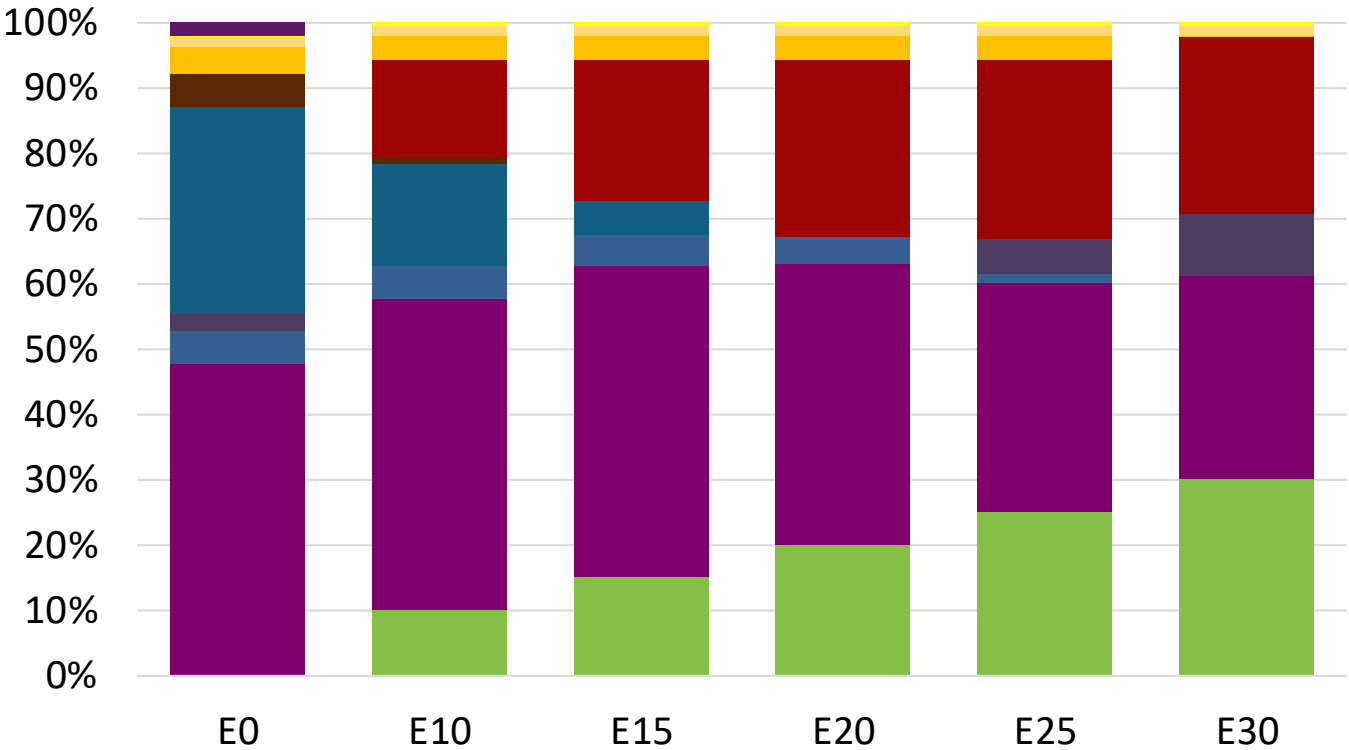
# Germany – Gasoline 95 – Constant Octane



Average CIF 2024  
Prices

|                  |          |          |          |          |          |          |
|------------------|----------|----------|----------|----------|----------|----------|
| Octane (RON)     | 95.0     | 95.0     | 95.0     | 95.0     | 95.1     | 95.6     |
| Price (US\$/gal) | \$ 2.272 | \$ 2.144 | \$ 2.083 | \$ 2.002 | \$ 1.880 | \$ 1.814 |

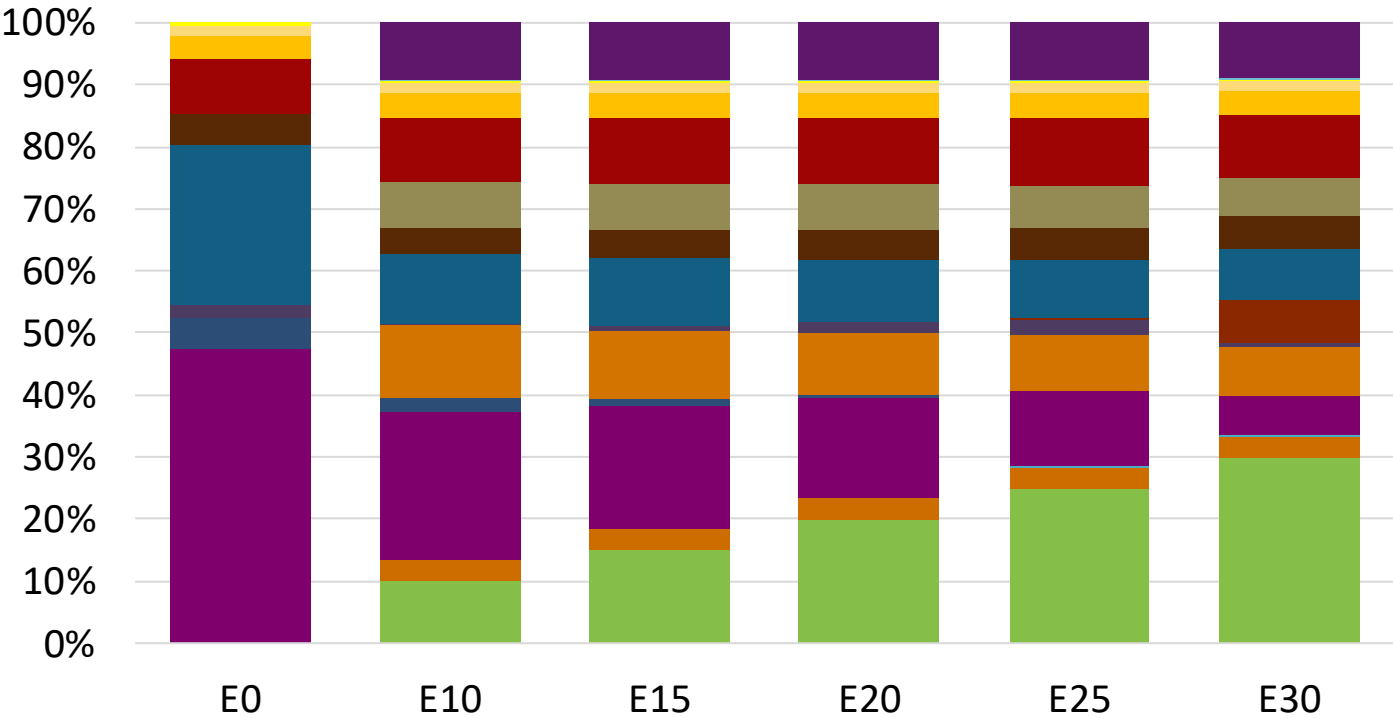
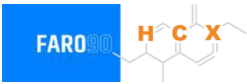
# Germany - Gasoline 98 - Constant Octane



Average CIF 2024  
Prices

|                  |          |          |          |          |          |          |
|------------------|----------|----------|----------|----------|----------|----------|
| Octane (RON)     | 98.0     | 98.0     | 98.0     | 98.0     | 98.0     | 98.0     |
| Price (US\$/gal) | \$ 2.347 | \$ 2.218 | \$ 2.146 | \$ 2.069 | \$ 1.992 | \$ 1.925 |

# Germany – Gasoline 95 – Increased Octane

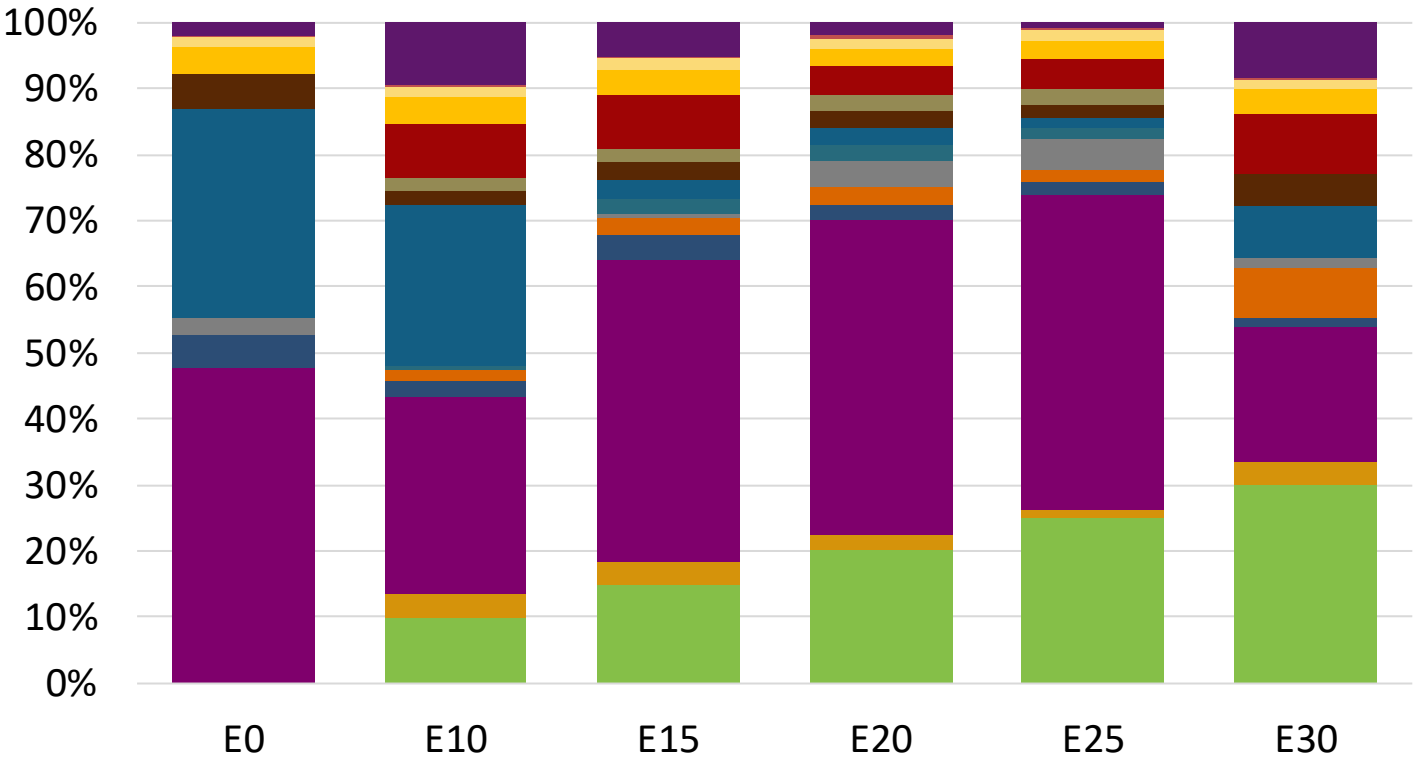
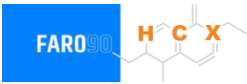


Average CIF 2024  
Prices

|                  |          |          |          |          |          |          |
|------------------|----------|----------|----------|----------|----------|----------|
| Octane (RON)     | 95.0     | 96.5     | 97.9     | 99.4     | 100.8    | 102.0    |
| Price (US\$/gal) | \$ 2.209 | \$ 2.209 | \$ 2.209 | \$ 2.209 | \$ 2.209 | \$ 2.209 |

Source: Faro90

# Germany – Gasoline 98 – Increased Octane



Average CIF 2024  
Prices

|                  |          |          |          |          |          |          |
|------------------|----------|----------|----------|----------|----------|----------|
| Octane (RON)     | 98.0     | 99.2     | 101.3    | 102.6    | 104.0    | 104.7    |
| Price (US\$/gal) | \$ 2.347 | \$ 2.347 | \$ 2.347 | \$ 2.347 | \$ 2.347 | \$ 2.347 |

Source: Faro90



# Vehicle Emission Impact for Ethanol Gasoline Blending

The model used in this analysis takes as a reference the [International Vehicle Emissions Model \(IVE\)](#).

The model uses the Base Emission Rates from IVE model, as well as its Adjustment Factors based on:

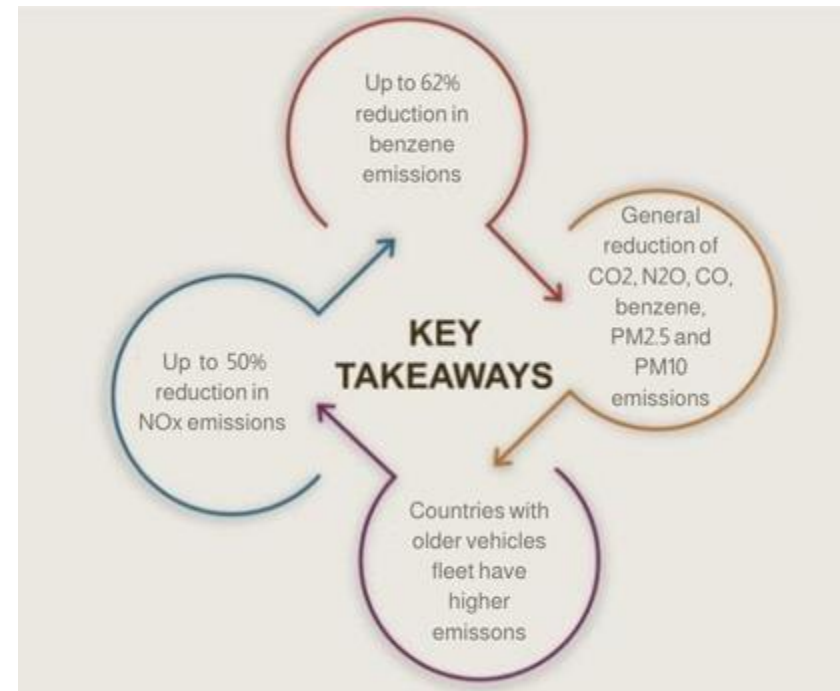
- Vehicle technology (cars, trucks, buses, motorcycles),
- Vehicle fleet average age,
- Average traveled distance per vehicle by country, as well as
- Geographical and climatic conditions (altitude, humidity, temperature).

Emissions of criteria pollutants, toxic pollutants, and greenhouse gases (GHG) were calculated and calibrated with emission inventories, using real gasoline quality data. The reduction rates for gasoline/ethanol blends were obtained from various sources (IPCC, US Grains, among others).

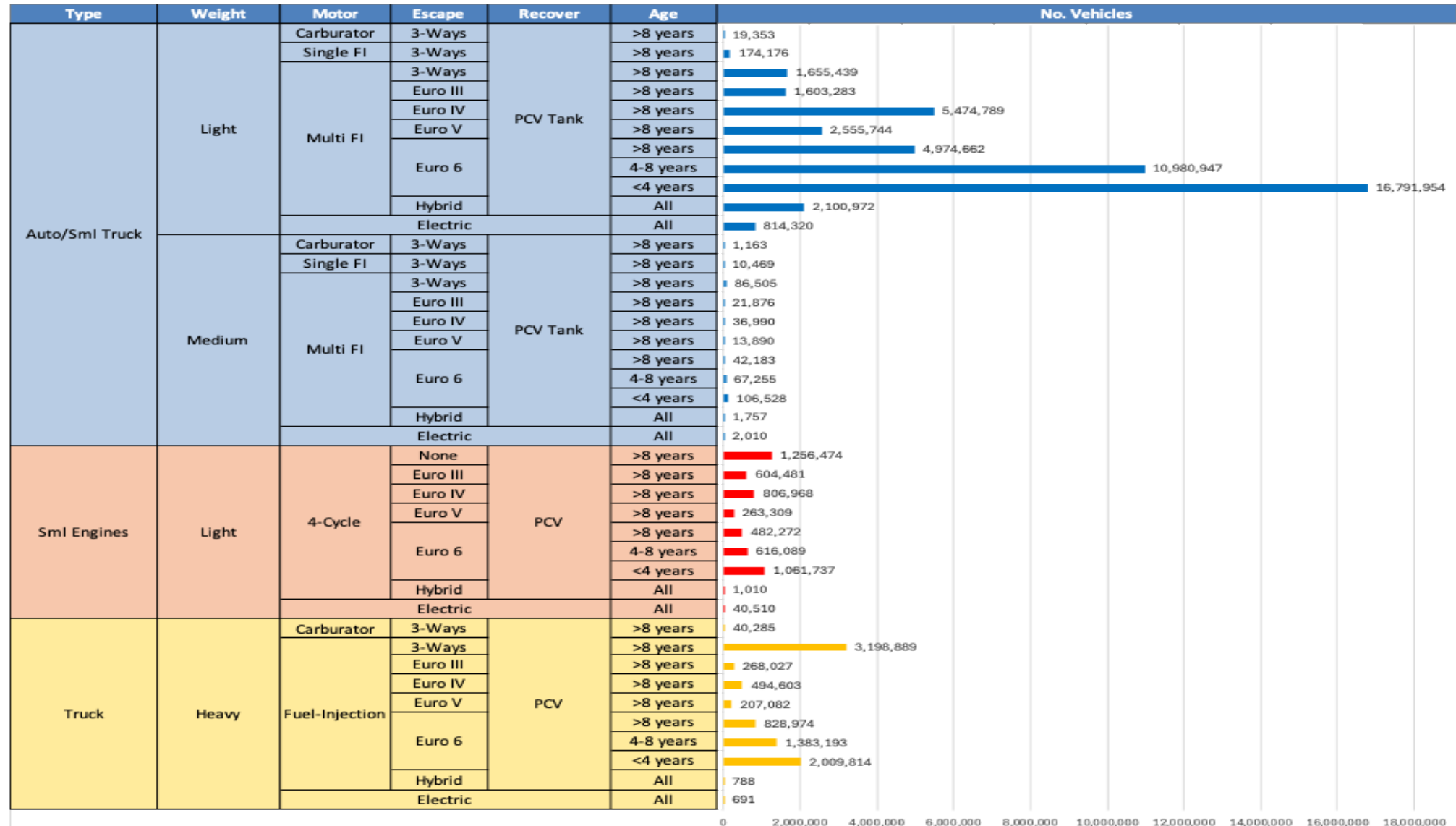
Emission estimations for different pollutants for gasoline and gasoline/ethanol blends (10%, 15%, 20%, 25% and 30% ethanol) were determined using the IVE Model. A comparison between the results and the European (Euro 6) requirements is made. Results are also compared with real emissions of the United States vehicle fleet\*.

\*Source: Bureau of transportation statistics.

## Main Results



# Gasoline Vehicle Fleet - Germany

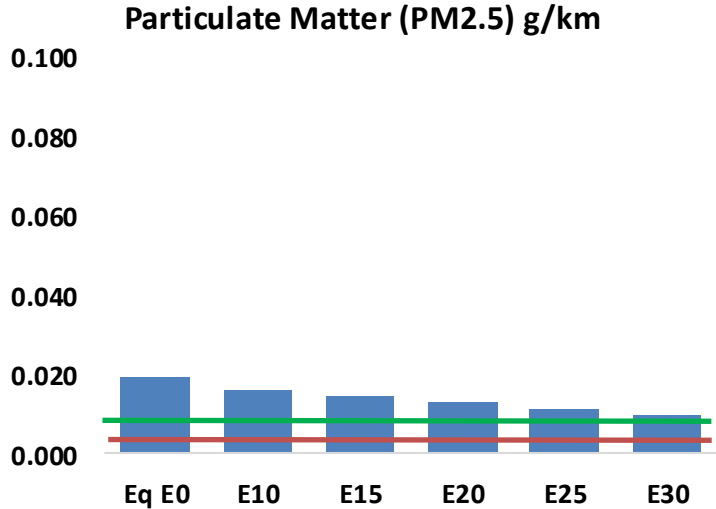
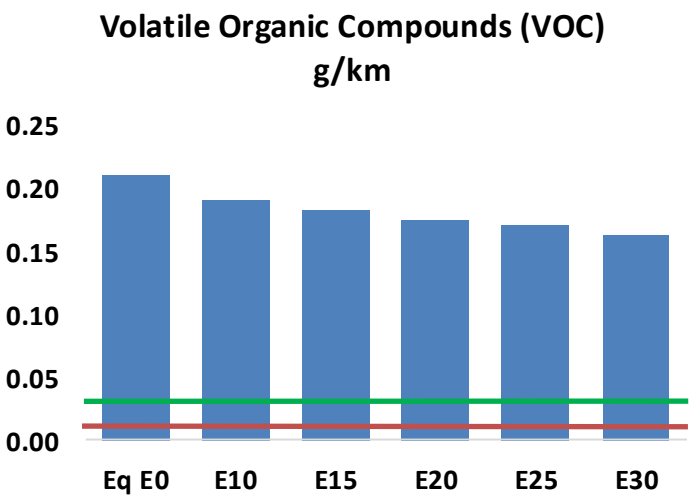
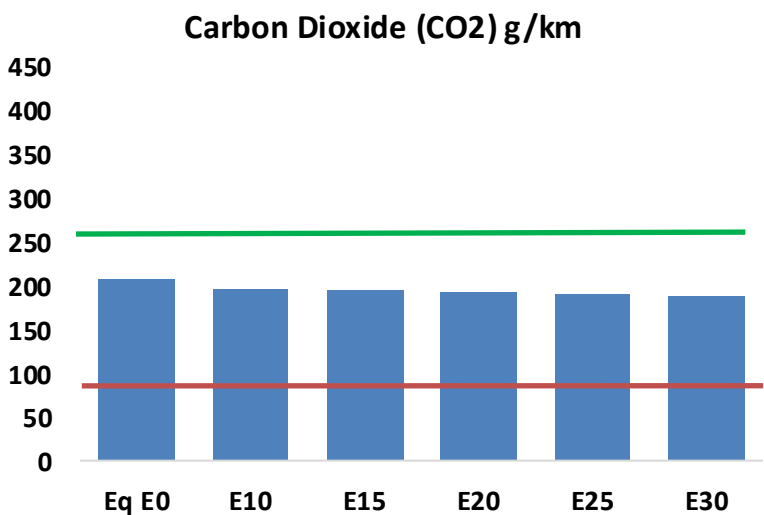
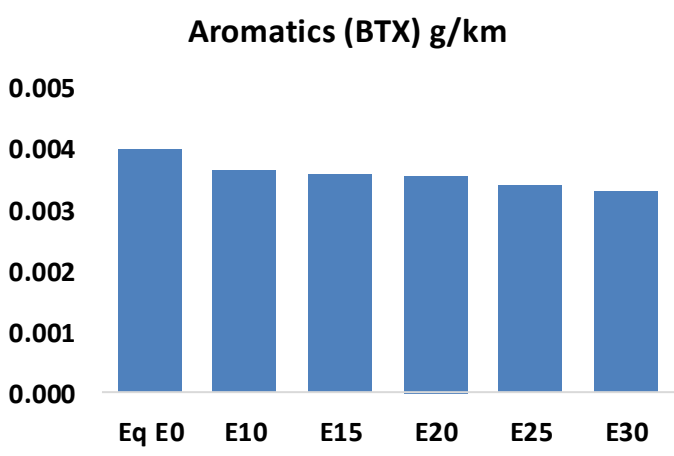
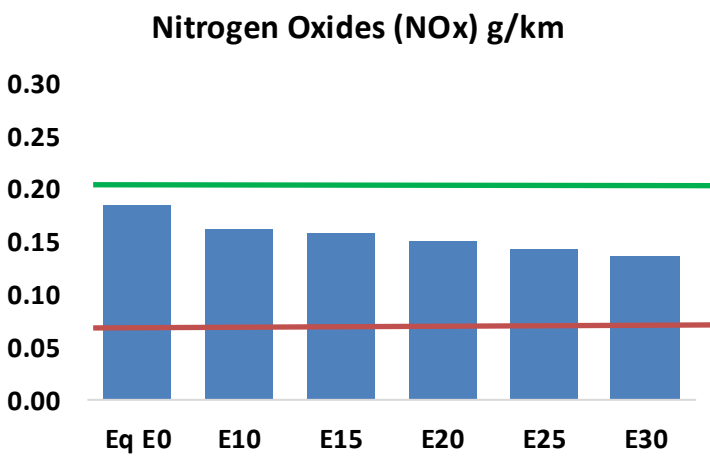
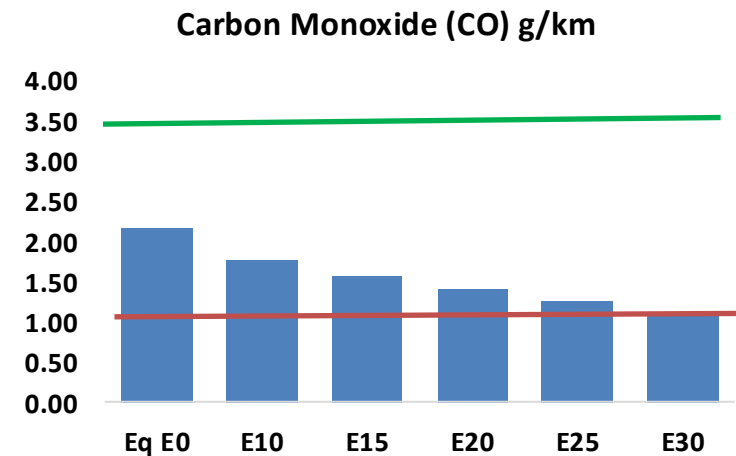
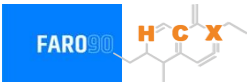


Vehicle Fleet: 61,101,463  
Gasoline Vehicle Fleet: 32,331,793

Source: Eurostat, ACEA

# Germany – Vehicular Emissions

TIER III BIN 125 United States  
Euro 6 Standard



# Germany – Vehicular Emissions

| Vehicular Fleet Emissions (kg/day) | Eq E0       | E5          | E10         | E15         | E20         | E25         | E30         | Reduction (%) |      |
|------------------------------------|-------------|-------------|-------------|-------------|-------------|-------------|-------------|---------------|------|
|                                    |             |             |             |             |             |             |             | E10           | E20  |
| CO                                 | 1,771,380   | 1,605,052   | 1,438,724   | 1,296,542   | 1,150,752   | 1,037,832   | 921,006     | -19%          | -35% |
| CO2                                | 169,687,313 | 165,323,925 | 161,202,948 | 157,961,920 | 156,358,133 | 155,431,613 | 152,566,562 | -5%           | -8%  |
| VOC                                | 172,226     | 163,701     | 155,175     | 149,108     | 143,427     | 139,186     | 133,075     | -10%          | -17% |
| NOx                                | 152,095     | 140,435     | 132,652     | 128,800     | 122,469     | 117,565     | 111,755     | -13%          | -19% |
| SOx                                | 1,887       | 1,744       | 1,602       | 1,481       | 1,365       | 1,240       | 1,108       | -15%          | -28% |
| NH3                                | 58,322      | 57,745      | 57,167      | 57,439      | 58,085      | 58,539      | 58,917      | -2%           | 0%   |
| N2O                                | 1,812       | 1,802       | 1,794       | 1,803       | 1,840       | 1,861       | 1,882       | -1%           | 2%   |
| CH4                                | 41,275      | 41,275      | 41,275      | 41,894      | 42,802      | 43,586      | 44,329      | 0%            | 4%   |
| Lead                               | -           | -           | -           | -           | -           | -           | -           | 0%            | 0%   |
| Butadiene                          | 1,660       | 1,582       | 1,504       | 1,450       | 1,400       | 1,362       | 1,308       | -9%           | -16% |
| Acetaldehyde                       | 3,370       | 3,868       | 5,675       | 8,642       | 11,774      | 13,454      | 15,898      | 68%           | 249% |
| Formaldehyde                       | 10,877      | 10,575      | 12,327      | 14,474      | 15,164      | 16,776      | 18,262      | 13%           | 39%  |
| Benzene                            | 3,279       | 3,151       | 2,986       | 2,939       | 2,914       | 2,787       | 2,697       | -9%           | -11% |
| PM2.5                              | 10,227      | 9,298       | 8,368       | 7,550       | 6,749       | 5,891       | 4,993       | -18%          | -34% |
| PM10                               | 5,724       | 5,203       | 4,683       | 4,225       | 3,777       | 3,297       | 2,794       | -18%          | -34% |

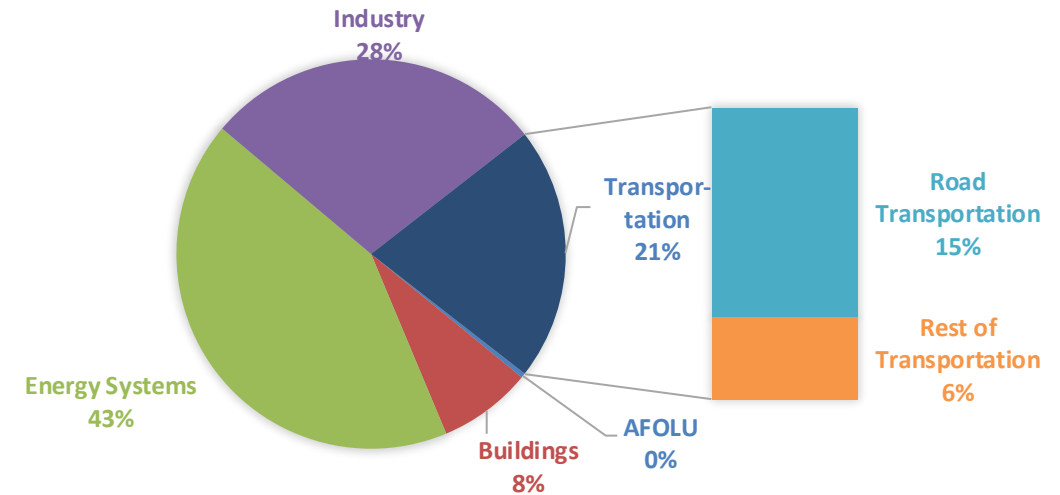
# Life Cycle GHG Emissions

The Global Agenda for Climate Action calls for countries, cities and regions, businesses, and members of civil society around the world to achieve a low-carbon and climate-resilient economies in support of the Paris Agreement. Road Transportation accounts for 15% of total GHG emissions (IPCC,2023).

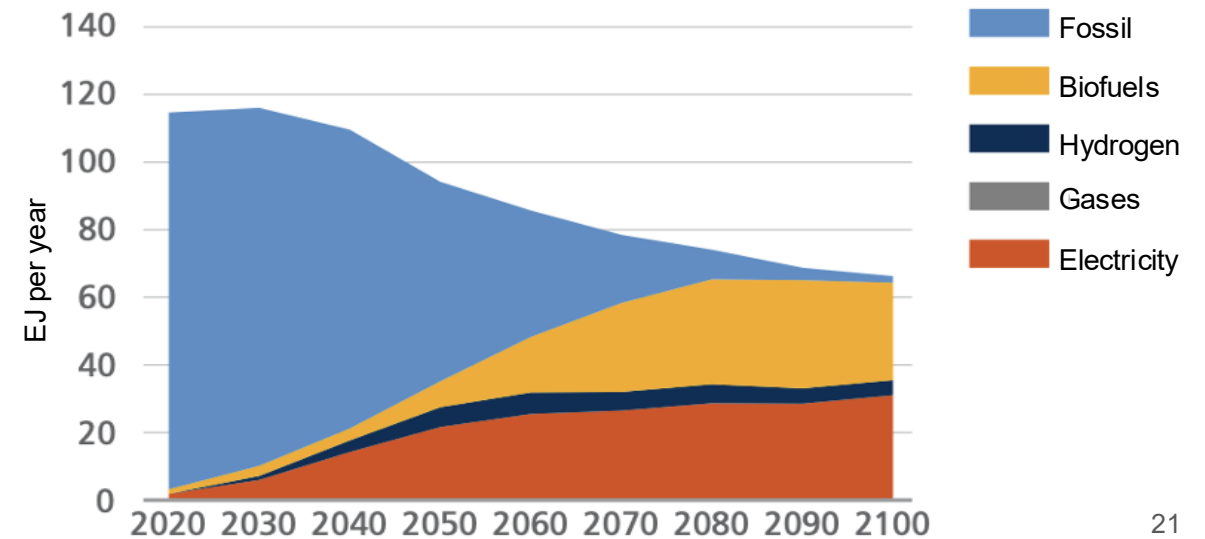
Ethanol gasoline blends are an internationally adopted practice that offer significant short-term benefits. Biofuels are an important solution to achieve reduction targets in the transportation sector.

Country GHG emission reductions are calculated using **RED II/III** and **GREET** life cycle emission methodologies and the **International Vehicular Emissions Model (IVE)**. Results are compared to the current country targets for 2035. Current 2024 country emission are those published by **IPCC**.

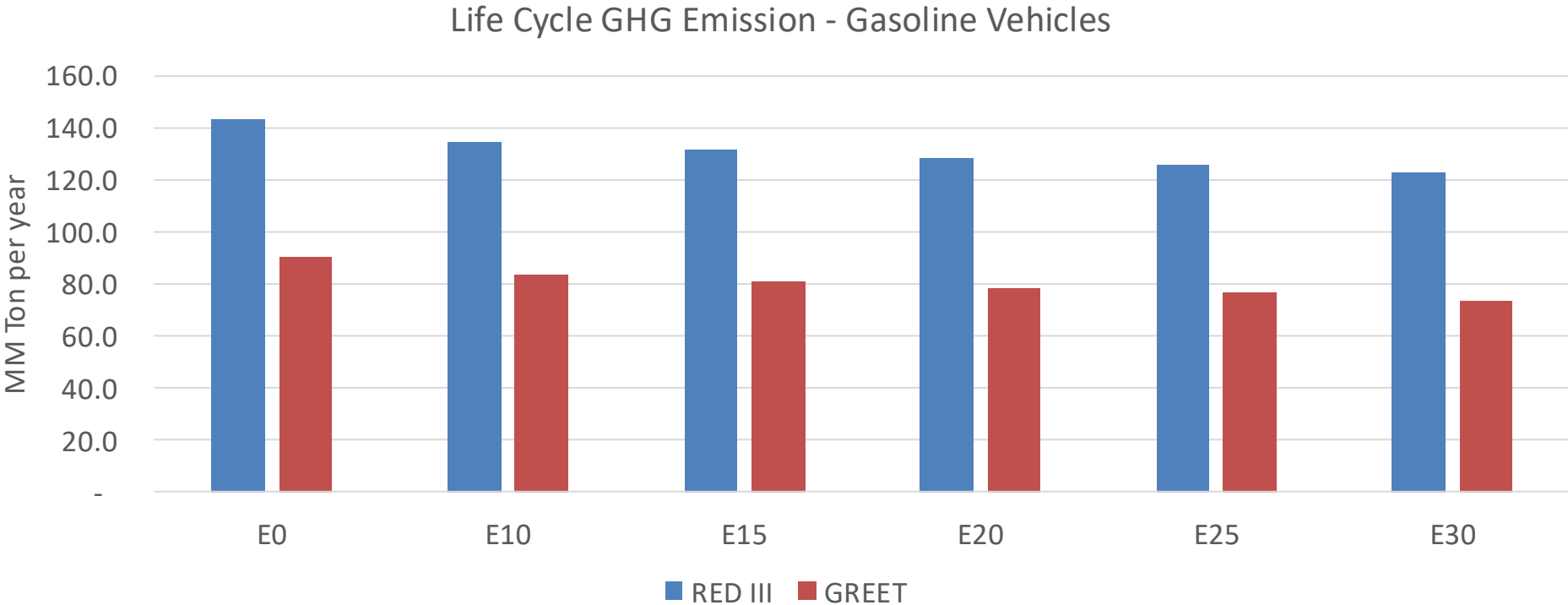
Current GHG Global Emissions Distribution



IPCC Transport Sector Sustainable Development Goals and climate policies Scenario, 2023



# Germany – Gasoline Vehicle Fleet Life Cycle GHG Emissions



|                                  |       |
|----------------------------------|-------|
| Country Emissions 2024 (MMT/año) | 717.1 |
| Target @ 2035 (MMT/year)         | 403.6 |
| Delta                            | 313.4 |

| vs E0      | E0   | E10  | E15   | E20   | E25   | E30   |
|------------|------|------|-------|-------|-------|-------|
| GREET 2024 | 0.0% | 8.0% | 11.0% | 13.6% | 15.8% | 19.0% |
| RED II/III | 0.0% | 5.9% | 8.1%  | 10.1% | 11.8% | 14.1% |

| Target     | E0   | E10  | E15  | E20  | E25  | E30  |
|------------|------|------|------|------|------|------|
| GREET 2024 | 0.0% | 2.3% | 3.2% | 3.9% | 4.6% | 5.5% |
| RED II/III | 0.0% | 2.7% | 3.7% | 4.6% | 5.4% | 6.4% |